Cross-Modal Hashing With Feature Semi-Interaction and Semantic Ranking for Remote Sensing Ship Image Retrieval

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***Abstract*— Cross-modal hashing plays a pivotal role in large-scale remote sensing (RS) ship image retrieval. RS ship images often exhibit similar overall appearance with subtle differences. Existing hashing methods typically employ feature non-interaction strategies to generate common hash codes, which may not effectively capture the correlations between cross-modal ship images to reduce intermodality discrepancies. To address this issue, we propose a novel cross-modal hashing approach based on feature semi-interaction and semantic ranking (FSISR) for RS ship image retrieval. Our FSISR approach not only captures intricate correlations between different ship image modalities, but also enables the construction of hash tables for large-scale retrieval. FSISR comprises a feature semi-interaction module and a semantic ranking objective function. The semi-interaction module utilizes clustering centers from one modality to learn the correlations between two modalities and generate robust shared representations. The objective function optimizes these representations in a common Hamming space, consisting of a shared semantic alignment loss and a margin-free ranking loss. The alignment loss employs a shared semantic layer to preserve label-level similarity, while the ranking loss incorporates hard examples to establish a margin-free loss that captures similarity ranking relationships. We evaluate the performance of our method on benchmark datasets and demonstrate its effectiveness for cross-modal RS ship image retrieval. https://github.com/sunyuxi/FSISR.**

***Index Terms*— Cross-modal remote sensing (RS) ship images, deep supervised hashing, learning to hash, multisource RS images, RS ship image retrieval.**

I. Introduction

With the advancement of Earth observation (EO) technologies, vast amounts of EO data are now generated and readily available [1], [2], [3], [4]. This has made it necessary to retrieve relevant images from available archives for various applications, such as satellite-based ocean monitoring systems [5], [6], which use image retrieval techniques to conduct large-scale early warnings and track maritime targets. Consequently, multiple remote sensing (RS) image retrieval tasks have been proposed, including image-to-image retrieval [5], [7], [8], [9], [10], [11], [12], [13], audio-to-image retrieval [14], [15], [16], [17], [18], and text-to-image retrieval [19], [20], [21], [22], [23]. This article focuses on the image-to-image retrieval task for cross-modal RS ship images. This task involves using a query image from one modality to search for similar images in a database from another modality.

Xiong et al. [5] first presented a publicly available dataset of cross-modal RS ship images, which includes a large collection of panchromatic (PAN) and multispectral (MS) images. They also proposed an interpretable fusion Siamese network (IFSN) to retrieve cross-modal RS ship images. IFSN extracts dense, high-dimensional features to match RS ship images across different modalities. However, when applying a large number of RS images, IFSN requires a lot of memory and a longer search time to find similar ship images, making it less effective. In contrast, cross-modal hashing methods, as described in [1], offer an effective solution for large-scale image-to-image retrieval to quickly and accurately identify similar ship images from a large collection of RS images. Cross-modal hashing aims to learn common hash codes between different image sources in Hamming space by reducing intermodal discrepancies. For example, Li et al. [24] proposed a source-invariant deep hashing network for PAN and MS images, which is based on a pairwise similarity constraint loss function. Xiong et al. [25] developed a cross-modal hashing

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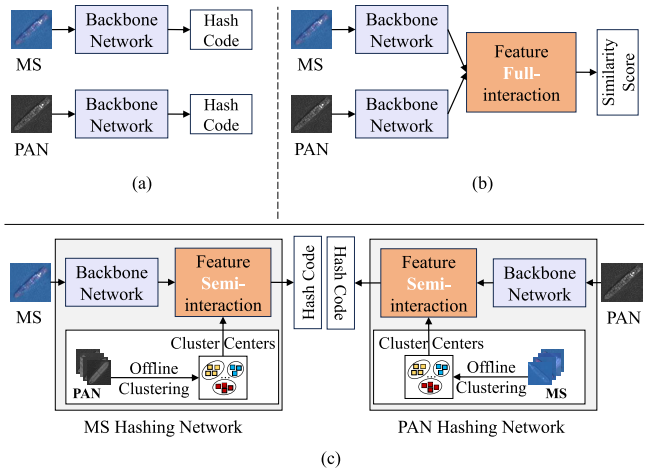


Fig. 1. (a) Existing hashing methods employ feature non-interaction to encode PAN and MS images into hash codes independently, which cannot effectively capture the correlation between the two modalities. (b) Existing non-hash retrieval techniques usually use feature full-interaction during the re-ranking stage to combine different modalities and generate similarity scores, which is not suitable for constructing hash tables. (c) Our proposed hashing approach utilizes cluster centers to implement feature semi-interaction, which can effectively learn semantic correlations and construct hash tables.

network with a triplet loss to reduce the discrepancies between MS and synthetic aperture radar (SAR) images. Although these approaches have demonstrated impressive results in cross-modal RS scenes, they encounter challenges in generating shared hash codes for ship object images due to their visual similarities with subtle differences in appearance.

However, existing cross-modal hashing techniques do not effectively facilitate feature interaction across different modalities to learn semantic correlations to distinguish ship images. Fig. 1(a) illustrates that current techniques, such as source-invariant deep hashing networks [24], independently encode MS and PAN images into hash codes, neglecting feature interaction. These networks struggle to capture correlations between different modalities, hindering the generation of robust shared hash codes. On the other hand, non-hash image retrieval methods, such as multimodal feature matching networks [19], employ full feature interaction strategies to combine different modalities and generate a similarity score, as shown in Fig. 1(b). Although these methods better understand the underlying semantic correlations, full feature interaction is impractical for hashing methods because query images are typically unavailable for database images during hash table construction.

To address these drawbacks, we propose a cross-modal hashing method with feature semi-interaction and semantic ranking for RS ship image retrieval, called FSISR. Our method effectively learns semantic correlations and fine-grained ranking structures across different modalities to distinguish RS ship images. Our method not only facilitates the construction of hash tables for large-scale cross-modal RS ship images but also demonstrates impressive retrieval results, effectively striking a balance between efficiency and performance. Our FSISR comprises feature semi-interaction modules and a semantic ranking objective function.

The semi-interaction module aims to fuse input images from one modality with clustering centers derived from the other modality. The two modalities do not interact directly but are connected through these clustering centers. This interaction is referred to as feature semi-interaction, as opposed to feature full-interaction for each pair of images from different modalities. This approach not only captures correlations between different modalities to create robust representations but also constructs separate hash tables for each modality. As depicted in Fig. 1(c), the PAN images interact with the clustering centers derived from the MS images. These centers are pre-computed offline using multiple MS images and only need to be computed once during the training of the hashing networks. The PAN hashing network only accepts PAN images as input to facilitate feature interactions and capture semantic correlations between the two modalities, resulting in a powerful shared representation. Similarly, the MS hashing network only takes MS images and follows a similar process to generate its own shared representation independently. Subsequently, these shared representations are transformed into unified hash codes in the Hamming space via a semantic ranking objective function. These hash codes can be utilized to build hash tables for efficient retrieval of large-scale PAN and MS images.

The semantic ranking objective function aims to align shared representations in the Hamming space, producing high-quality hash codes. This function consists of two components: a shared semantic alignment loss and a margin-free ranking loss. The alignment loss utilizes a shared semantic layer to reduce discrepancies and preserve label-level similarity between cross-modal ship images. The ranking loss employs the nearest neighbor hard samples from each sample to learn the ranking structure of cross-modal RS ship images. Unlike traditional ranking losses with fixed margins, our approach uses hard examples to implement a margin-free ranking loss, which effectively captures similarity ranking relationships and improves retrieval performance. Our method applies this objective function to generate unified hash codes for cross-modal RS ship images. Comprehensive experiments have been conducted to demonstrate its effectiveness in cross-modal RS ship image retrieval.

The contributions of this article are summarized as follows.

1) A novel cross-modal hashing retrieval framework based on FSISR is designed to learn powerful shared hash codes for cross-modal RS ship image retrieval.
2) To learn semantic correlations among cross-modal RS ship images, we design a novel feature semi-interaction module within hashing networks. This module not only captures the correlations between different ship object modalities to generate robust shared representations, but also enables the construction of hash tables for quick retrieval.
3) To align shared representations in the Hamming space, we propose a semantic ranking objective function. This function employs a shared semantic alignment loss and a margin-free ranking loss to effectively preserve label-level semantic similarity and ranking relations of cross-modal ship images. Experimental results demonstrate that our method significantly improves the retrieval performance of cross-modal ship images.

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The rest of this article is structured as follows. In Section II, we provide a brief overview of the related work. Section III offers a comprehensive explanation of our method. Section IV presents the experimental results in detail. Finally, Section V concludes the article with a summary.

II. Related Work

This section reviews unimodal and cross-modal hashing methods.

A. Unimodal Hashing Methods

Unimodal hashing methods [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [35], [36], [37], [38], [39], [40], [41], [42], [43] aim to acquire binary features in the Hamming space that maintain the similarity between RS images from the same satellite sensor. RS images often present intricate visual features, posing a challenge for models to comprehensively understand the visual information [44], [45]. Leveraging the powerful image representation capabilities of deep networks, deep learning has emerged as a prevalent approach for handling RS images [46]. For example, Li et al. [26] developed a deep hashing network that uses the semantic similarity of image pairs to learn hash functions for RS images. Song et al. [36] combined the similarity of image pairs and the semantic details of class labels to generate more discriminative hash codes. Liu et al. [37] unified the classification and retrieval of RS images into a unified framework, which uses class labels as semantic cues to further improve retrieval performance. Han et al. [38] developed a lightweight deep hashing model with a class center loss to generate hash codes that can identify similar patches in large RS scenes. Chen et al. [39] proposed deep saliency smoothing hashing to address drone image retrieval, which learns hash codes by combining saliency capture, distribution smoothing, and the incorporation of global and local features. Han et al. [40] focused on encrypted hashing for localization tasks, designing an encrypted patch generator to encrypt image patches with minor changes in visual appearance but specific changes in hash code. These approaches usually produce good retrieval performance when labeled images are abundant.

The effectiveness of deep unimodal hashing methods can be limited when there is a lack of labeled images. To address this issue, Tang et al. [41] proposed a meta-learning-based supervised hashing approach that effectively trains hashing networks using a small set of labeled training samples. Hou et al. [42] used labeled data from a source domain to create hash functions for unlabeled data in a target domain using unsupervised domain adaptation algorithms. Fernandez-Beltran et al. [35] used a probabilistic graph to generate discriminative hash codes, where the graph is constructed based on a similarity matrix between RS images. Sun et al. [43] proposed a soft-pseudo-label-based unsupervised deep hashing method for RS image retrieval, which employs soft pseudo-labels to accurately describe the global similarity between intercluster images by binarized vectors. However, these techniques are only applicable to unisource RS images and cannot handle cross-source RS images. In practical scenarios, such as all-weather rescue [47], [48], the use of cross-source RS images is essential. The direct application of unimodal methods for learning cross-source hashing functions is not possible [25]. Therefore, there is an increasing interest in the development of cross-modal hashing methods.

B. Cross-Modal Hashing Methods

Cross-modal hashing techniques [24], [25], [49], [50], [51], [52], [53], [54], [55], [56], [57], [58] are employed to reduce the discrepancies between cross-modal RS data by learning nonlinear hash functions and encoding them into hash codes in a shared Hamming space. These methods have been applied to various cross-source RS image retrieval tasks. Li et al. [24] proposed a source-invariant deep hashing network with a pairwise similarity-constrained loss function for PAN and MS image retrieval. Xiong et al. [25] developed a cross-modal hashing network with triplet loss to reduce the differences between MS and SAR images. Sun et al. [57] proposed a multisensor fusion module and an explicit semantic-preserving objective function to reduce the discrepancies between very-high-resolution (VHR) and SAR images. Chaudhuri et al. [11] created a cross-modal sketch-image dataset and proposed a novel framework for zero-shot intermodal data retrieval of RS data, which utilizes an encoder–decoder structure with cross-triplet and cross-projection loss functions to improve overlapping between the two data domains. Fang et al. [56] trained two hashing models for sketches and RS images, using a pair of jointly constructed hash codebooks based on interclass pairwise disperse cues and intraclass pairwise balance cues. Chaudhuri et al. [58] introduced a label-deficit zero-shot learning strategy with a Siamese network for cross-source RS image retrieval.

Cross-modal hashing is a popular approach used for text-to-image and audio-to-image retrieval tasks. For example, Jiang and Li [53] developed an end-to-end hashing approach that combines feature learning and hash-code learning into a framework. This method does not require relaxation and can directly learn discrete hash codes. Cao et al. [54] designed a pairwise focal loss based on the exponential distribution to generate compact and highly concentrated hash codes and to reduce the large heterogeneity across different modalities. Yuan et al. [19] proposed an asymmetric multimodal feature matching network to enable text-to-image matching. This network includes a multiscale visual self-attention module to extract important features from RS images, which are then used to direct the representation learning of text. Tang et al. [23] designed an information-interacting enhancement module to simultaneously model the inner relationships between RS images and texts to improve the performance of image retrieval. Chen et al. [14] proposed a deep quadruple-based hashing technique that uses a quadruple construction hard module to learn the semantic similarity between audio and RS images. Chen et al. [18] developed a fine-aligned discriminative hashing approach to capture relevant information from RS images and audio. These methods usually yield promising results in

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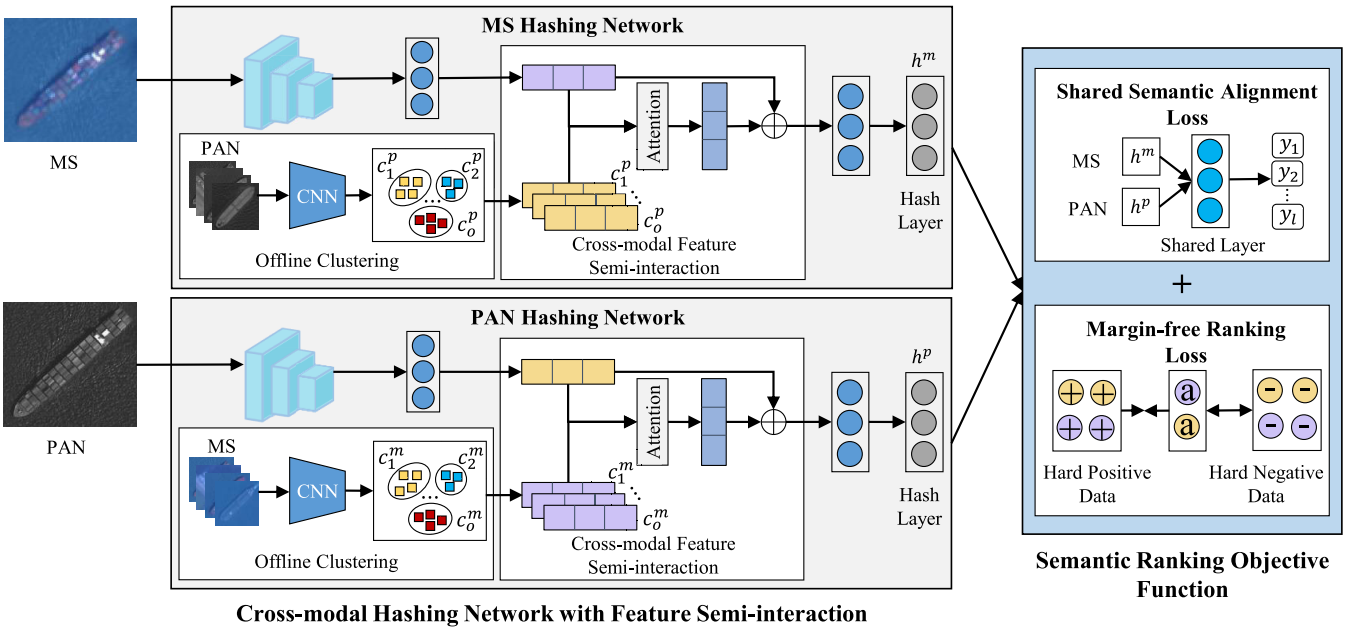


Fig. 2. Overview of the proposed FSISR method based on FSISR for cross-modal RS ship image retrieval. The method is composed of a cross-modal hashing network with feature semi-interaction and a semantic ranking objective function. The hash codes generated by the MS hashing network and the PAN hashing network are denoted as h^m and h^p, respectively.

image-to-image, text-to-image, and audio-to-image retrieval tasks.

Existing cross-modal hashing techniques mainly study cross-modal RS scenes rather than cross-modal RS ship objects. Xiong et al. [5] first collected several publicly accessible PAN and MS images to investigate cross-modal RS ship image retrieval. When existing cross-modal hashing techniques were applied to these images, they usually did not yield satisfactory results since they were not able to effectively promote feature interaction across different modalities to reduce intermodality discrepancies. Consequently, we propose a cross-modal hashing method for ship object retrieval, which incorporates a feature semi-interaction module and a semantic ranking objective function.

III. Method

In the cross-modal RS ship image retrieval task, we are provided with a dataset consisting of n instances $\mathcal{X} = \{(x_i^m, x_i^p, y_i)\}_{i=1}^n$, where x_i^m and x_i^p are MS and PAN images, respectively, and $y_i \in \{1, 2, \ldots, l\}$ is the class label of the ith instance, with l being the total number of classes. The goal of cross-modal hashing for ship image retrieval is to encode x^m and x^p into K-bit hash codes by learning two nonlinear hashing functions: $f_{x^m} : x^m \mapsto h^m \in \{-1, 1\}^K$ and $f_{x^p} : x^p \mapsto h^p \in \{-1, 1\}^K$. The hashing functions need to ensure that h^m and h^p have a small Hamming distance when x^m and x^p belong to the same class, and a large Hamming distance when they belong to different classes.

A. Overview

As illustrated in Fig. 2, we propose a novel cross-modal hashing method based on feature semi-interaction and semantic ranking for RS ship image retrieval, called FSISR, which comprises two main components: a cross-modal hashing network with feature semi-interaction and a semantic ranking objective function.

The cross-modal hashing network consists of two separate hashing networks for MS and PAN images. Each network is composed of a feature extraction network, an offline clustering module, a cross-modal feature semi-interaction module, and a hash layer. The feature extraction network is similar to that of [24] and [57]. The goal of the hashing networks is to generate hash codes for their respective image sources. The semantic ranking objective function is composed of a shared semantic alignment loss and a margin-free ranking loss. The alignment loss preserves label-level semantic similarity by using a shared semantic layer, while the ranking loss learns the similarity ranking relationships. The objective function optimizes the hashing networks to generate high-quality hash codes in a common Hamming space.

B. Cross-Modal Hashing Network With Feature Semi-Interaction

Cross-modal hashing methods aim to reduce the discrepancies between different modalities and generate hash codes in a unified Hamming space. Existing approaches either employ feature non-interaction strategies or feature full-interaction strategies to learn shared representations, but neither of them can effectively balance efficiency and performance in cross-modal retrieval of RS ship images. For example, hashing networks with feature non-interaction use independent networks to learn hashing functions, which can be used to construct hash tables for large-scale cross-modal RS ship image retrieval, but they are not able to effectively capture the correlations between different modalities. On the other hand, image retrieval based on full feature interaction concatenates two modalities of data and applies self-attention to yield rich interactions, which can understand the semantic

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correlations between different modalities and achieve better results. However, they cannot be used to construct hash tables for large-scale cross-modal RS ship image retrieval. Therefore, we propose a cross-modal hashing network with feature semi-interaction to achieve good performance and enable the construction of hash tables for quick retrieval.

The proposed cross-modal network uses a feature semi-interaction approach to combine input images from one modality with clustering centers generated from the other modality. These clustering centers, serving as feature vectors, are generated based on the visual features of the second modality. Unlike direct interaction between the two modalities, this connection is established through clustering centers, constituting what we term as feature semi-interaction. In this semi-interaction, the visual features of input images act as Queries, while clustering centers function as Keys and Values. Leveraging the attention mechanism, we compute attention weights based on input images, enabling the calculation of the weighted average of clustering centers. This implementation facilitates interaction between input images and cluster centers.

The proposed cross-modal network consists of two independent hashing networks with feature semi-interaction: the MS hashing network and the PAN hashing network. Both networks have the same structure, but the MS hashing network works with MS images, while the PAN hashing network operates on PAN images. We will mainly focus on explaining the design of the MS hashing network, which is composed of four components: a feature extraction network, an offline clustering module, a cross-modal feature semi-interaction module, and a hash layer.

As illustrated in Fig. 2, the feature extraction network in the MS hashing network encodes MS ship images into high-dimensional features through convolutional layers and fully connected layers. This is expressed as follows:

$$z_i^m = f_m\left(x_i^m, W_z^m\right) \in \mathbb{R}^d \tag{1}$$

where x_i^m is the ith MS image, z_i^m is the high-dimensional feature of x_i^m, and f_m is the feature extraction network with the parameter W_z^m.

The offline clustering module in the MS hashing network utilizes convolutional neural networks (CNNs) to extract visual features from PAN images. These features are then put through K-means clustering to generate cluster centers. The process is outlined as follows:

$$\left\{c_i^p\right\}_{i=1}^o = \text{Cluster}\left(g_p\left(x^p, W_c^p\right)\right) \in \mathbb{R}^{o\times d} \tag{2}$$

where x^p represents a set of PAN images, g_p denotes the CNN with the parameter W_c^p, and $\{c_i^p\}_{i=1}^o$ represents the resulting cluster centers. This approach utilizes a sample of images from the PAN collection to generate cluster centers, enabling us to capture the data distribution of PAN images with a limited number of cluster centers. When a MS image interacts with PAN images, it does not directly fuse with a single PAN image, but rather with the cluster centers that were previously generated. This approach eliminates the need for MS and PAN images simultaneously when constructing hash tables for one modality. Additionally, MS images can interact with the cluster centers of PAN images to create robust representations. We refer to the process as semi-interaction between two-modality images, rather than full-interaction. This feature semi-interaction provides a balance between the efficiency and effectiveness of representing MS images within hashing networks.

The cross-modal feature semi-interaction module in the MS hashing network employs an attention mechanism to fuse the visual features of an MS image with cluster centers derived from PAN images. As depicted in Fig. 2, to generate the fused features, we first pre-process the original feature of the MS image, denoted as $z_i^m \in \mathbb{R}^d$, and the cluster centers of PAN images, denoted as $c_i^p \in \mathbb{R}^d$, to produce normalized features. This process is expressed as follows:

$$\hat{z}_i^m = \frac{z_i^m(j)}{\sqrt{\sum_{t=1}^d z_i^m(t)^2}} \tag{3}$$

$$\hat{c}_i^p = \frac{c_i^p(j)}{\sqrt{\sum_{t=1}^d c_i^p(t)^2}} \tag{4}$$

where $\hat{z}_i^m$ and $\hat{c}_i^p$ are the generated normalized features, and $z_i^m(j)$ and $c_i^p(j)$ are the jth element of z_i^m and c_i^p, respectively. Next, we use an attention mechanism to enable interaction between the normalized features, generating a combined representation. This attention mechanism is described as follows:

$$s_{ij} = \frac{\exp\left(\hat{z}_i^m \hat{c}_j^p\right)}{\sum_{t=1}^o \exp\left(\hat{z}_i^m \hat{c}_t^p\right)}$$
$$e_i^m = \sum_{j=1}^o s_{ij}\hat{c}_j^p \tag{5}$$

where s_{ij} is an attention weight and e_i^m is a fused feature for the MS image and cluster centers of PAN images. The fused feature is obtained by taking a weighted sum of PAN image cluster centers using the computed attention weights s_{ij}. Finally, we concatenate the fused features e_i^m with the normalized feature $\hat{z}_i^m$ to generate the ultimate feature representation of the MS image, which is defined as follows:

$$\hat{e}_i^m = \text{ReLU}\left(\text{MLP}\left(e_i^m\right) \parallel \hat{z}_i^m\right) \tag{6}$$

where $\hat{e}_i^m$ represents the final feature vector for the MS image, obtained through concatenation after applying the rectified linear unit (ReLU) activation function and a multilayer perceptron (MLP) transformation.

The hash layer maps the high-dimensional feature $\hat{e}_i^m$ into a low-dimensional representation and then compresses it into a compact hash code using a sign function, as formulated by

$$h_i^m = \text{sign}\left(g_h\left(\hat{e}_i^m, W_h^m\right)\right), \quad h_i^m \in \{-1, 1\}^K \tag{7}$$

with

$$\text{sign}(x) = \begin{cases} 1, & \text{if } x \geq 0 \\ -1, & \text{if } x < 0 \end{cases} \tag{8}$$

where g_h is a nonlinear mapping network with parameter W_h^m and is responsible for transforming $\hat{e}_i^m$ into a hash-like feature. This network is usually composed of fully connected layers, normalization layers, and activation functions. The sign$(\cdot)$

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function is an element-wise sign function that returns 1 if the element is positive and -1 otherwise. The resulting h_i^m is the desired hash code of the MS image.

The PAN and MS hashing networks both have the same components, such as a feature extraction network, an offline clustering module, a cross-modal feature semi-interaction module, and a hash layer. However, they do not share parameters and accept different input images and cluster centers. For example, the MS hashing network processes MS images with PAN image cluster centers, while the PAN hashing network works with PAN images and MS image cluster centers. The PAN hashing network follows the same operating process as the MS hashing network to generate its hash code $h_i^p \in \{-1, 1\}^K$, which is defined as follows:

$$h_i^p = \operatorname{sign}\left(g_h\left(\hat{e}_i^p, W_h^p\right)\right), \quad h_i^p \in \{-1, 1\}^K \tag{9}$$

where $\hat{e}_i^p$ represents the final feature vector for a PAN image. To train the two hashing networks, a semantic ranking objective function is designed to reduce intermodality differences and generate high-quality hash codes in a common Hamming space.

C. Semantic Ranking Objective Function

Cross-modal hashing aims to generate binarized representations of RS ship images from multiple sources in a common Hamming space. Cross-modal RS ship objects typically exhibit a high degree of visual similarity with subtle differences, making it challenging to develop shared hash codes that achieve excellent ranking performance. To generate high-quality hash codes, we use class labels and ranking relationships of cross-modal RS ship images to design a semantic ranking objective function. This objective function effectively preserves semantic similarity and maintains ranking relationships across different modalities, comprising a shared semantic alignment loss and a margin-free ranking loss.

1) Shared Semantic Alignment Loss: The MS and PAN hashing networks generate robust hash-like features for cross-modal RS ship images. To keep the semantic information of these features in the Hamming space, we utilize class labels and softmax functions to optimize the hashing networks. Additionally, different modalities of RS ship images often exhibit significant heterogeneity gaps. To reduce this gap, we incorporate a shared semantic layer with softmax functions to implement a shared semantic alignment loss. The layer is represented by a learnable matrix. This loss effectively preserves semantic similarity across different modalities and is defined as follows:

$$\begin{aligned}\mathcal{L}_{\text{ssa}} = &\frac{1}{n}\sum_{i=1}^{n} -\log\frac{\exp\left(u_y h_i^m\right)}{\sum_{j=1}^{l}\exp\left(u_j h_i^m\right)} \\ &+ \frac{1}{n}\sum_{i=1}^{n} -\log\frac{\exp\left(u_y h_i^p\right)}{\sum_{j=1}^{l}\exp\left(u_j h_i^p\right)} \\ \text{s.t.} \quad & h_i^m = \operatorname{sign}\left(g_h\left(\hat{e}_i^m, W_h^m\right)\right) \in \{-1, 1\}^K \\ & h_i^p = \operatorname{sign}\left(g_h\left(\hat{e}_i^p, W_h^p\right)\right) \in \{-1, 1\}^K \end{aligned} \tag{10}$$

where h_i^m and h_i^p are the hash codes generated by the MS and PAN hashing networks, and $u_j \in \mathbb{R}^K$ is a feature vector of the learnable matrix. In particular, u_y is the yth feature vector of the matrix.

The loss function is designed to bring h_i^m and h_i^p into a shared semantic space to learn high-quality common hash codes. However, the sign$(\cdot)$ function used to generate h_i^m and h_i^p is non-differentiable at zero, which prevents the loss function from optimizing and updating the hashing networks through backpropagation algorithms. To address this problem, we replace the sign$(\cdot)$ function with the tanh$(\cdot)$ function and introduce a hash quantization loss to constrain the hash codes. Inspired by Zhu et al. [59], this hash quantization loss is defined as follows:

$$\begin{aligned}\mathcal{L}_q = &\frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{K}\log\cosh\left(\left|h_i^m(j)\right| - 1\right) \\ &+ \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{K}\log\cosh\left(\left|h_i^p(j)\right| - 1\right)\end{aligned} \tag{11}$$

where $h_i^m(j)$ and $h_i^p(j)$ are the jth elements of h_i^m and h_i^p, respectively, and $|\cdot|$ calculates the absolute value. This loss encourages the hash codes to be as close as possible to the values $\{-1, 1\}$.

By using (11), we can simplify (10) to get the final shared semantic alignment loss, which is expressed as

$$\begin{aligned}\mathcal{L}_{\text{ssa}} = &\frac{1}{n}\sum_{i=1}^{n} -\log\frac{\exp\left(u_y h_i^m\right)}{\sum_{j=1}^{l}\exp\left(u_j h_i^m\right)} \\ &+ \frac{1}{n}\sum_{i=1}^{n} -\log\frac{\exp\left(u_y h_i^p\right)}{\sum_{j=1}^{l}\exp\left(u_j h_i^p\right)} \\ &+ \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{K}\log\cosh\left(|h_i^m(j)| - 1\right) \\ &+ \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{K}\log\cosh\left(|h_i^p(j)| - 1\right).\end{aligned} \tag{12}$$

2) Margin-Free Ranking Loss: Cross-modal hashing retrieval utilizes hash codes to match similar RS ship images. To ensure that similarity ranking relationships between different modalities are preserved within the Hamming space, we propose a novel margin-free ranking loss function. This loss function is designed to directly learn these ranking relationships using hard example mining strategies. Traditional ranking loss functions (such as triplet loss) have a fixed margin parameter that is used to distinguish between similar and dissimilar samples. This makes it difficult to adjust to complex data distributions. However, our ranking loss obtains the closest hard positive and negative samples from each sample to be able to learn the ranking structure of cross-modal RS ship images in a more flexible way.

The formulation of the ranking loss is as follows:

$$\begin{aligned}\mathcal{L}_{\text{mfr}} = &\frac{1}{n}\sum_{i=1}^{n}\max\left(0, \|h_i^m - h_i^{p+}\|_2 - \|h_i^m - h_i^{p-}\|_2\right) \\ &+ \frac{1}{n}\sum_{i=1}^{n}\max\left(0, \|h_i^p - h_i^{m+}\|_2 - \|h_i^p - h_i^{m-}\|_2\right)\end{aligned} \tag{13}$$

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where $\|\cdot\|_2$ is the Euclidean distance, h_i^m is the hash code of an MS image, and h_i^p is the hash code of a PAN image. h_i^{p+} and h_i^{p-} are the hash codes of the PAN images corresponding to the positive and negative data of h_i^m, respectively. Similarly, h_i^{m+} and h_i^{m-} are the hash codes of the MS images corresponding to the positive and negative data of h_i^p, respectively. The loss is calculated by summing the maximum difference between the hash codes of the hard positive and hard negative data for each image.

The ranking loss involves determining positive images by calculating the last-k most similar images for each image in the training set, and negative images by selecting the top-k most dissimilar images. For example, we calculate the similarities between an MS image and all PAN images in the training dataset, and arrange them in descending order. The last-k PAN images with the same label as the MS image are considered hard positive examples, while the top-k images with different labels from the MS image are considered hard negative examples. Our ranking loss uses these hard examples (instead of a fixed margin value) to optimize the networks, allowing us to effectively train hashing networks to learn the intricate neighborhood structure of cross-modal RS ship images, and thus generate highly discriminative hash codes.

According to (12) and (13), we derive the semantic ranking objective function as follows:

$$\mathcal{L} = \mathcal{L}_{\text{ssa}} + \lambda_0 \mathcal{L}_{\text{mfr}} \tag{14}$$

where λ_0 is a hyperparameter. By optimizing $\mathcal{L}$, we can effectively train the hashing networks to learn robust hash codes for cross-modal RS ship image retrieval.

IV. Experiments

A. Experimental Setup

1) Dataset: To demonstrate the effectiveness of FSISR, we conducted a comparative analysis of our method against state-of-the-art approaches on MRSSID [5] and VAIS [60]. MRSSID [5] is a publicly available dataset suitable for cross-modal RS ship image retrieval. This dataset comprises six distinct categories of ship images, including destroyers, littoral combat ships, combat boats, bulk carriers, container ships, and oil tankers. It includes PAN and MS ship images, all acquired simultaneously within the same geographic area. The PAN ship images have a spatial resolution of 1 m, while the MS ship images have a resolution of 4 m, containing red, green, blue, and near-infrared bands. We followed the protocol in [5] to split this dataset into training and testing sets. VAIS [60] is a ship image dataset that includes visible and infrared ship images. The dataset comprises six ship categories: merchant, sailing, passenger, medium, tug, and small. VAIS is incorporated as a supplementary dataset for validation purposes. We followed the protocol in [60] to split this dataset into training and testing sets.

2) Implementation Details: Our FSISR approach was implemented using the PyTorch [61] framework. We used an Adam optimizer [62] with decay rates $\beta_1 = 0.9$ and $\beta_2 = 0.999$. The learning rate was initially set to 1.0×10^{-4} and decreased by 0.9 every ten epochs. The training batch size was fixed at 128, and the total number of epochs was 200. The hyperparameter λ_0 was set to 1.0. Parameter analysis is discussed in Section IV-D.

3) Baselines: We compared FSISR with nine classical or state-of-the-art supervised cross-modal hashing methods. These methods encompass four traditional shallow approaches (CMFH [51], IMH [49], LSSH [50], SCM [52]), and five recently introduced deep learning methods (CMHH [54], DCMH [53], SIDHCNN [24], DSMHN [55], MsEspH [57]).

1) *CMFH* [51] employs collective matrix factorization with latent factor models from different modalities to learn unified hash codes.
2) *IMH* [49] explores correlations across various media types from different data sources and incorporates a linear regression model to learn efficient hashing functions.
3) *LSSH* [50] employs sparse coding and matrix factorization to learn latent semantic features from different modalities. It adopts an iterative strategy to efficiently obtain optimal solutions.
4) *SCM* [52] is a cross-modal hash method based on semantic correlation maximization that integrates semantic labels into the hashing learning procedure.
5) *CMHH* [54] utilizes an exponential distribution to design a pairwise focal loss, reducing the heterogeneity among various modalities and generating compact common hash codes.
6) *DCMH* [53] is an end-to-end hashing approach that combines feature learning and hash-code learning to directly learn discrete hash codes without relaxation constraints.
7) *SIDHCNN* [24] is a source-invariant deep hashing network with a pairwise similarity-constrained loss function for PAN and MS image retrieval.
8) *DSMHN* [55] leverages feature representation learning, intermodality similarity preservation, semantic label preservation, and hash function learning to generate compact and high-quality hash codes.
9) *MsEspH* [57] incorporates a multisensor fusion module and an explicit semantic-preserving objective function to minimize discrepancies between very high resolution (VHR) and synthetic aperture radar (SAR) images, resulting in discriminative hash codes within a common Hamming space.

Following the methods in [24] and [57], shallow hashing methods employ deep features extracted by a CNN model as input, while deep hashing methods use raw image pixels to train hashing networks. To ensure a fair and consistent comparison, we adopted the same backbone network for all deep hashing methods.

4) Evaluation Protocol: We utilized several evaluation metrics to evaluate the retrieval performance of all methods [11], [16], [24], [25], [38], [57], such as mean average precision (MAP), precision–recall curves, and Top-N precision ($P@N$) curves. The definitions of these metrics are as follows.

1) The MAP is a metric used to evaluate the performance of information retrieval systems across a set of query images. It is calculated by taking the mean of the average

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TABLE I
COMPARISON OF MAP OF HAMMING RANKING FOR DIFFERENT BITS ON CROSS-MODAL RS SHIP IMAGE RETRIEVAL

Architecture	Method	MS→PAN					PAN→MS				
		16 bits	24 bits	32 bits	48 bits	64 bits	16 bits	24 bits	32 bits	48 bits	64 bits
Shallow	CMFH	0.1264	0.1207	0.1250	0.1319	0.1273	0.1391	0.1326	0.1475	0.1540	0.1510
	IMH	0.1358	0.1332	0.1386	0.1422	0.1388	0.1295	0.1250	0.1274	0.1322	0.1323
	LSSH	0.1310	0.1331	0.1362	0.1406	0.1380	0.1240	0.1295	0.1292	0.1354	0.1375
	SCM	0.3052	0.2882	0.2935	0.2946	0.2926	0.4146	0.4186	0.4352	0.4288	0.4177
Deep	CMHH	0.6123	0.5333	0.5354	0.5065	0.4931	0.7133	0.6390	0.6409	0.6089	0.5992
	DCMH	0.5600	0.5593	0.5657	0.5749	0.5750	0.6950	0.6998	0.6829	0.6533	0.6398
	SIDHCNN	0.6010	0.5969	0.5838	0.5872	0.5418	0.7033	0.6924	0.7119	0.6875	0.6715
	DSMHN	0.6399	0.6482	0.6243	0.6494	0.6369	0.7938	0.7972	0.7918	0.7975	0.7801
	MsEspH	0.6681	0.6691	0.6687	0.6673	0.6492	0.7890	0.7793	0.8024	0.7834	0.7946
	FSISR	**0.7151**	**0.7253**	**0.6843**	**0.7127**	**0.6948**	**0.8212**	**0.8204**	**0.8058**	**0.8313**	**0.8164**

precision (AP) scores for each query. The formula for MAP is expressed as follows:

$$\mathrm{MAP} = \frac{1}{Q}\sum_{q=1}^{Q}\mathrm{AP}(x_q) \tag{15}$$

where Q is the total number of queries and $\mathrm{AP}(x_q)$ is the AP score for the specific query image x_q. To calculate the AP for each query, the following formula is used:

$$\mathrm{AP}(x_q) = \frac{1}{R}\sum_{r=1}^{R} P(x_q, r)\theta(r) \tag{16}$$

where R is the count of ground-truth images in the database relevant to each query. $P(x_q, r)$ is the precision of the rth retrieved image, which is sorted based on their Hamming distance from the query image x_q in ascending order. The indicator function $\theta(r)$ is used to determine whether the rth retrieved image shares at least one common label with the query x_q. If there is a match, $\theta(r)$ is set to 1; otherwise, it is set to 0. In this article, R is set to 86 when calculating the MAP for all methods.

2) The precision–recall curve illustrates the varying precision values at different recall values. These precision and recall values are calculated using different Hamming radii during the search process. A larger area under the precision–recall curve indicates better retrieval performance. The curve provides useful information on the balance between precision and recall, enabling us to adjust retrieval systems for optimal results. It is an essential tool for evaluating information retrieval algorithms.

3) The $P@N$ metric evaluates the precision of the top-N recalled images. It is calculated by dividing the number of images similar to the query image (denoted as k) by the total number of retrieved images (denoted as N), which is defined as follows:

$$P@N = \frac{k}{N}. \tag{17}$$

This metric shows the proportion of images similar to the query image among the top-N retrieved images. In other words, it reveals how many images among the top-N retrieved images are similar to the query image.

B. Experimental Results

1) Comparison With Baseline Methods: We performed a comprehensive quantitative comparison of our FSISR method with existing cross-modal hashing methods on the MRSSID dataset with hash code lengths ranging from 16 to 64. The MAP scores of all methods are reported in Table I for cross-modal RS ship image retrieval. In this table, "Shallow" denotes traditional shallow hashing methods that leverage deep features, while "Deep" indicates deep hashing methods based on neural networks.

Table I shows that shallow hashing methods exhibit the worst performance, even though they take deep features as input. This is because shallow methods rely on hand-crafted kernel functions to learn image representations and hashing functions separately, which does not provide effective feedback for generating more discriminative hash codes. In contrast, all deep hashing methods outperform shallow hashing methods, as deep methods can utilize CNNs to jointly learn image representations and hashing functions, thereby automatically capturing relevant features from cross-modal data. For example, when compared with SCM on the 24-bit hash code, MsEspH shows remarkable absolute improvements of 38.09% and 36.07% in the "MS→PAN" and "PAN→MS" tasks, respectively. Furthermore, our FSISR method outperforms all baseline methods. For example, when compared with MsEspH in the "MS→PAN" task, FSISR exhibits relative improvements of 7.03% and 8.40% for 16-bit and 24-bit hash codes, respectively. Similarly, in the "PAN→MS" task, FSISR also achieves the improvements of 4.08% and 5.27% for 16-bit and 24-bit hash codes, respectively. This is because our FSISR method employs feature semi-interaction to more effectively learn correlations between different modalities compared with MsEspH. In summary, FSISR outperforms all baseline approaches across various hash code lengths in cross-modal RS ship image retrieval tasks.

The precision–recall curves for different methods across various hash codes are presented in Fig. 3. These curves demonstrate that our FSISR consistently outperforms the other methods when applied to MS and PAN ship images, as seen from the larger area under the curve. For example, in Fig. 3(a) for the "MS→PAN" task on 16-bit hash codes, the

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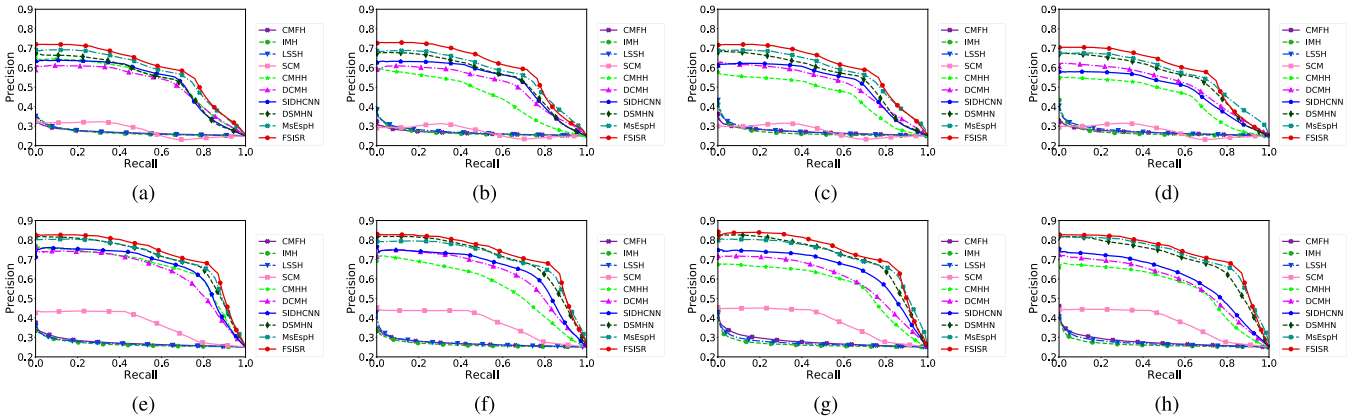


Fig. 3. Precision–recall curves obtained for different methods on the MRSSID dataset. (a) MS→PAN@16 bits. (b) MS→PAN@24 bits. (c) MS→PAN@48 bits. (d) MS→PAN@64 bits. (e) PAN→MS@16 bits. (f) PAN→MS@24 bits. (g) PAN→MS@48 bits. (h) PAN→MS@64 bits.

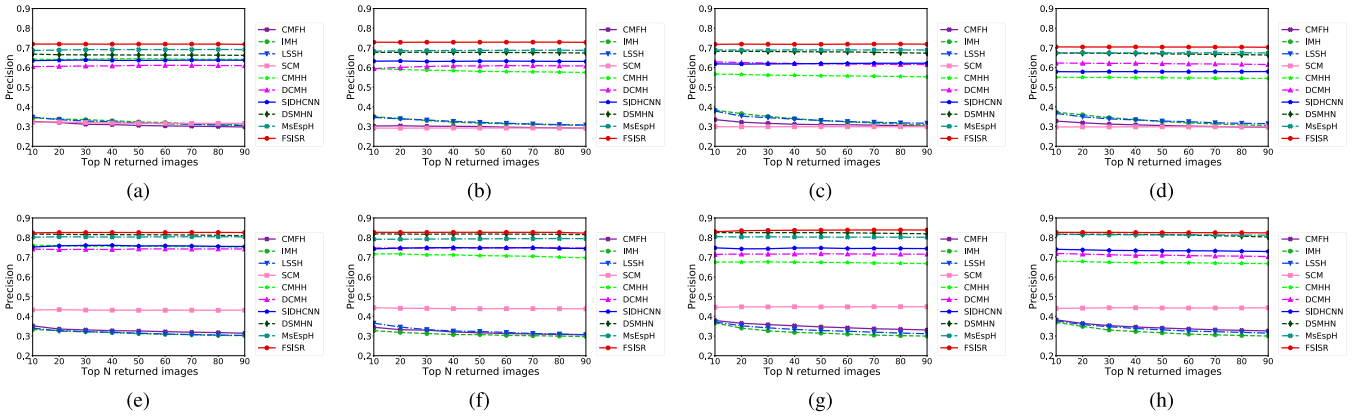


Fig. 4. $P@N$ curves on the MRSSID dataset. (a) MS→PAN@16 bits. (b) MS→PAN@24 bits. (c) MS→PAN@48 bits. (d) MS→PAN@64 bits. (e) PAN→MS@16 bits. (f) PAN→MS@24 bits. (g) PAN→MS@48 bits. (h) PAN→MS@64 bits.

precision–recall curve of FSISR has the largest area, indicating its superior performance compared with the other methods. Additionally, it is noteworthy that deep hashing methods generally achieve higher precision levels than shallow hashing methods at various recall values. For example, in Fig. 3(f) for the "PAN→MS" task, there is a significant precision gap between FSISR and SCM when using 24-bit hash codes. In summary, FSISR outperforms all the baseline methods in different hash codes and tasks.

By analyzing the Top-N precision curves of various hashing techniques under different N values in Fig. 4, we can gain a better understanding of our experimental results. We can see that FSISR outperforms other methods in terms of precision for MS and PAN ship images. For example, in the "MS→PAN" task performed using 16-bit hash codes shown in Fig. 4(a), FSISR exhibits the highest Top-N precision curve among all methods. Furthermore, we observe that FSISR maintains high precision across different N values, demonstrating its robustness in retrieval tasks. This is because FSISR adopts a semantic ranking objective to effectively learn ranking relationships and semantic similarity between different modalities of ship images, thus improving image retrieval performance. Therefore, these experimental results not only demonstrate the outstanding performance of FSISR in retrieval tasks but also emphasize its significant advantages when recalling different numbers of MS and PAN ship images.

We also present the MAP results of FSISR, DSMHN, and MsEspH for each class label in Table II. We observe that FSISR outperforms other methods across most class labels for various bit lengths. For example, compared to DSMHN in the "MS→PAN" task for bulk carriers, FSISR achieves increments of 23.95% and 6.70% for 16-bit and 24-bit hash codes, respectively. Furthermore, compared to MsEspH in the "MS→PAN" task for bulk carriers, FSISR achieves improvements of 16.76% and 9.78% for 16-bit and 24-bit hash codes, respectively. This is because our FSISR method utilizes feature semi-interaction strategies to effectively capture correlations between MS and PAN ship images, thus reducing intermodality differences and generating high-quality common hash codes. In conclusion, FSISR demonstrates a superior retrieval performance compared with DSMHN and MsEspH across different bit lengths.

We also conducted experiments on VAIS and presented the MAP results of FSISR alongside state-of-the-art methods in Table III. "IR" and "VIS" denote infrared and visible ship images, respectively. Our FSISR outperforms other methods in ship image retrieval. For example, compared to MsEspH with a 16-bit hash code, FSISR shows significant improvements of 7.15% and 0.27% for the "IR→VIS" and "VIS→IR" tasks, respectively. This superior performance can be attributed to our method's incorporation of a feature semi-interaction module and a semantic ranking objective function. These components facilitate learning semantic correlations across different

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TABLE II
MAP RESULTS FOR FSISR, DSMHN, AND MSESPH ARE PRESENTED FOR EACH CLASS LABEL IN MS AND PAN SHIP IMAGES

Task	Class Label	DSMHN				MsEspH				FSISR			
		16 bits	24 bits	32 bits	48 bits	16 bits	24 bits	32 bits	48 bits	16 bits	24 bits	32 bits	48 bits
MS→PAN	destroyers	0.5734	0.6923	0.5137	0.5370	0.6671	0.7399	0.7584	0.7336	0.7692	0.8453	0.6923	0.7652
	littoral combat ships	0.6216	0.6733	0.5808	0.6500	0.7212	0.7408	0.7273	0.6837	0.8056	0.7500	0.8013	0.7494
	combat boats	0.5789	0.5065	0.5835	0.5392	0.5673	0.6093	0.5070	0.5985	0.5282	0.6223	0.6452	0.5831
	bulk carriers	0.5604	0.6120	0.5418	0.5732	0.5949	0.5948	0.6200	0.5783	0.6946	0.6530	0.6577	0.6213
	container ships	0.7022	0.7064	0.6704	0.6282	0.7333	0.7289	0.7333	0.7609	0.7313	0.7231	0.6896	0.7794
	oil tankers	0.7278	0.6955	0.7266	0.7876	0.7380	0.7198	0.7215	0.7345	0.7625	0.8228	0.6957	0.8106
	all	0.6399	0.6482	0.6243	0.6494	0.6681	0.6691	0.6687	0.6673	**0.7151**	**0.7253**	**0.6843**	**0.7127**
PAN→MS	destroyers	0.7765	0.6648	0.5825	0.5896	0.7773	0.7614	0.7637	0.7214	0.7692	0.8462	0.8462	0.8462
	littoral combat ships	0.7598	0.8402	0.8078	0.6942	0.8097	0.8327	0.8370	0.8101	0.8333	0.8333	0.8151	0.8056
	combat boats	0.6526	0.6631	0.6324	0.6826	0.6544	0.6030	0.6762	0.5859	0.6464	0.6667	0.5718	0.6991
	bulk carriers	0.7590	0.8017	0.7615	0.7827	0.7205	0.7499	0.7399	0.7399	0.7960	0.7665	0.7825	0.8320
	container ships	0.7342	0.7011	0.7365	0.7261	0.7776	0.7595	0.7891	0.7832	0.8239	0.8350	0.7860	0.7586
	oil tankers	0.9131	0.8796	0.9143	0.9238	0.9084	0.8648	0.9124	0.8919	0.9034	0.9160	0.9074	0.9100
	all	0.7938	0.7972	0.7918	0.7975	0.7890	0.7793	0.8024	0.7834	**0.8212**	**0.8204**	**0.8058**	**0.8313**

TABLE III
COMPARISON OF MAP ON VAIS FOR CROSS-MODAL SHIP IMAGE RETRIEVAL

Architecture	Method	IR→VIS				VIS→IR			
		16 bits	24 bits	48 bits	64 bits	16 bits	24 bits	48 bits	64 bits
Shallow	CMFH	0.2424	0.2310	0.2445	0.2352	0.3879	0.4007	0.4444	0.4347
	IMH	0.3282	0.3199	0.3091	0.2936	0.3967	0.3831	0.4029	0.3932
	LSSH	0.2534	0.2975	0.3187	0.3267	0.2916	0.3243	0.3502	0.3710
	SCM	0.5393	0.5495	0.5497	0.5497	0.7378	0.7207	0.7207	0.6997
Deep	CMHH	0.9380	0.9182	0.8279	0.8377	0.9754	0.9253	0.8357	0.9021
	DCMH	0.8710	0.8895	0.8814	0.8739	0.9234	0.9540	0.9383	0.9282
	SIDHCNN	0.9184	0.9009	0.8938	0.8857	0.9567	0.9678	0.9745	0.9615
	DSMHN	0.9249	0.9448	0.9269	0.9137	0.9874	0.9908	0.9874	0.9841
	MsEspH	0.9167	0.9330	0.9301	0.9306	0.9883	0.9885	0.9876	0.9820
	FSISR	**0.9822**	**0.9820**	**0.9884**	**0.9910**	**0.9910**	**0.9889**	0.9868	**0.9910**

modalities and reduce intermodality discrepancies, leading to enhanced retrieval performance. Furthermore, deep hashing methods consistently outperform shallow hashing methods. For example, when compared with SCM in the "VIS→IR" task, FSISR exhibits absolute improvements of 25.32% and 26.82% for 16-bit and 24-bit hash codes, respectively. Similarly, in the "IR→VIS" task, FSISR achieves improvements of 44.29% and 43.25% for 16-bit and 24-bit hash codes, respectively. This superiority is attributed to the end-to-end learning of image representations and hashing functions by deep hashing methods, which generate higher-quality hash codes compared to shallow hashing methods. In summary, FSISR achieves the highest performance among all methods on VAIS.

2) Visualization of Retrieval Results: In this section, we present the retrieval results of FSISR and two baselines, namely MsEspH and DCMH, to showcase their performance. Fig. 5 displays the top 10 retrieved RS ship images from each method. The "Query" column indicates the query images from different modalities, with the first two queries being MS images, and the last two queries being PAN images. Red squares in the displayed images denote incorrect results. From these retrieval results, we can see that FSISR retrieves more visually and semantically similar ship images compared to other methods. For example, in the "MS→PAN" task focusing on oil tankers, FSISR excels in returning images that are semantically consistent with the query. In contrast, MsEspH and DCMH provide visually similar images but often from different semantic categories. FSISR can effectively distinguish RS ship images that may be confusing, such as those of oil tankers. This capability is derived from FSISR's utilization of feature semi-interaction to learn the correlations between MS and PAN images, thereby effectively distinguishing these confusing ship images.

C. Ablation Study

We conducted an ablation study to evaluate the impact of feature semi-interaction and margin-free ranking loss in our proposed approach. In the study, "w/o FSI" and "w/o MFRank" denote scenarios where feature semi-interaction and

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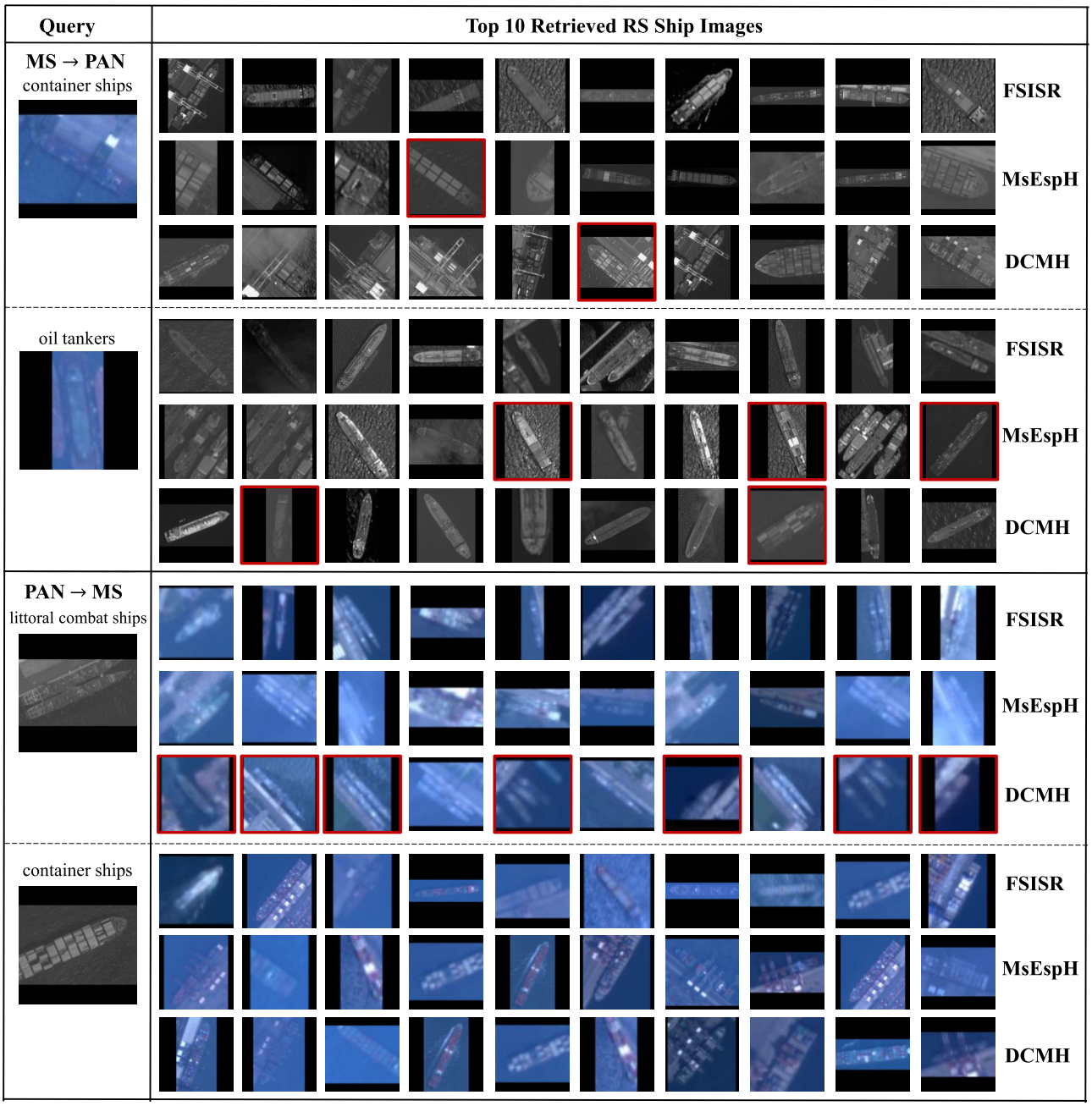


Fig. 5. Top-10 retrieved RS ship images by FSISR, MsEspH, and DCMH for MS→PAN and PAN→MS tasks.

TABLE IV
ABLATION TEST RESULTS IN TERMS OF MAP BY DISCARDING FEATURE SEMI-INTERACTION AND MARGIN-FREE RANKING LOSS

Task	Method	16 bits	24 bits	48 bits
MS→PAN	w/o FSI	0.7073	0.6642	0.6860
	w/o MFRank	0.7113	0.7075	0.6902
	FSISR	0.7151	0.7253	0.7127
PAN→MS	w/o FSI	0.8084	0.8062	0.7961
	w/o MFRank	0.8151	0.8111	0.8037
	FSISR	0.8212	0.8204	0.8313

margin-free ranking loss are removed, respectively. The results of the ablation tests for the "MS→PAN" and "PAN→MS" tasks are presented in Table IV.

From the obtained results, we observe that feature semi-interaction significantly enhances the retrieval performance of FSISR by capturing correlations between MS and PAN images. For example, in the "MS→PAN" task, FSISR improves by 1.10% and 3.89% for 16-bit and 48-bit hash codes compared to the "w/o FSI" method. Similarly, in the "PAN→MS" task, FSISR exhibits improvements of 1.58% and 4.42%. Furthermore, the margin-free ranking loss contributes to FSISR by learning the fine-grained ranking structure of cross-modal data. In the "MS→PAN" task, FSISR achieves increments of 2.52% and 3.26% for 24-bit and 48-bit hash codes compared to "w/o MFRank." Similarly, in the "PAN→MS" task, FSISR demonstrates improvements of 1.15% and 3.43%. Precision–recall curves of the ablation results are depicted in Fig. 6. The precision–recall curve of

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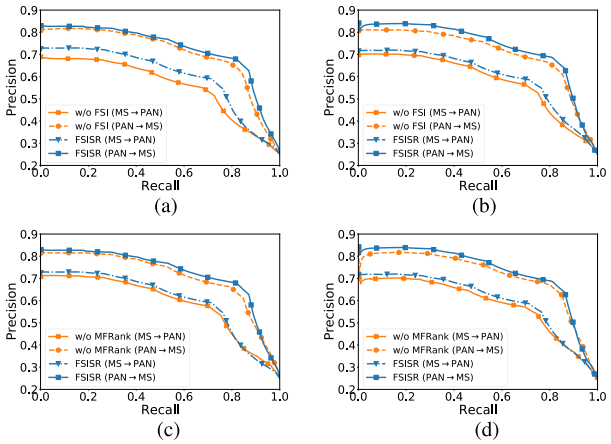


Fig. 6. Precision–recall curves of w/o FSI, w/o MFRank, and FSISR. (a) Precision–recall curves of w/o FSI and FSISR on 24 bits. (b) Precision–recall curves of w/o FSI and FSISR on 48 bits. (c) Precision–recall curves of w/o MFRank and FSISR on 24 bits. (d) Precision–recall curves of w/o MFRank and FSISR on 48 bits.

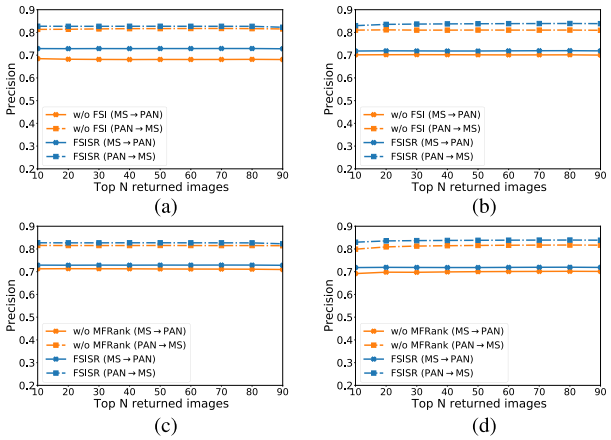


Fig. 7. Top-N precision curves of w/o FSI, w/o MFRank and FSISR. (a) Top-N precision curves of w/o FSI and FSISR on 24 bits. (b) Top-N precision curves of w/o FSI and FSISR on 48 bits. (c) Top-N precision curves of w/o MFRank and FSISR on 24 bits. (d) Top-N precision curves of w/o MFRank and FSISR on 48 bits.

FSISR has a larger area in the "MS→PAN" and "PAN→MS" tasks, illustrating that FSISR consistently outperforms both the "w/o FSI" and "w/o MFRank" methods. Fig. 7 displays Top-N precision curves of the ablation results, with the precision curve of FSISR consistently higher, showcasing its robustness across various N values in retrieval tasks.

In conclusion, FSISR effectively utilizes feature semi-interaction and margin-free ranking loss, capturing intermodality correlations and learning the fine-grained ranking structure of cross-modal data. This results in a significant enhancement in retrieval performance.

D. Parameter Analysis

We conducted experiments to evaluate the effect of the hyperparameter λ_0 on FSISR with 24-bit hash codes, and the results are shown in Fig. 8. λ_0 is used to balance shared semantic alignment loss and margin-free ranking loss, thus controlling the importance of ranking loss in the objective function. As shown in Fig. 8, when λ_0 is equal to 1, FSISR achieves the best results for both "MS→PAN" and "PAN→MS" tasks. To analyze the influence of λ_0, we varied its values and measured the MAP of FSISR in the tasks "MS→PAN" and "PAN→MS". As depicted in Fig. 8, the MAP score changes distinctly as λ_0 varies for two tasks. Initially, the MAP score increases as the value of λ_0 increases, followed by a gradual decrease. The best results are obtained when λ_0 is equal to 1. When λ_0 is close to 1, FSISR effectively preserves label-level semantic similarity and ranking relationships between cross-modal ship images. However, when λ_0 is too large, FSISR tends to overly emphasize ranking relations at the expense of semantic similarity. In conclusion, the hyperparameter λ_0 allows us to adjust the balance between semantic similarity and ranking relations to learn high-quality hash codes within a common Hamming space.

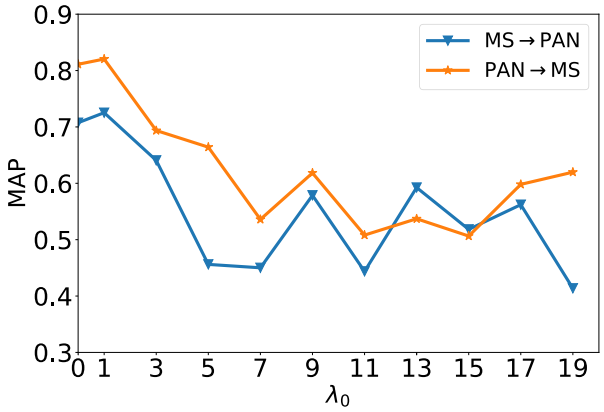


Fig. 8. Sensitivity analysis of λ_0 on 24 bits.

V. Conclusion

This study tackled feature interaction across different modalities in RS ship image retrieval. To effectively learn semantic correlations and align cross-modal ship images in the Hamming space, we introduce a novel cross-modal hashing approach based on feature semi-interaction and semantic ranking, named FSISR. The proposed method consists of two primary components: a cross-modal hashing network with feature semi-interaction and a semantic ranking objective function. The cross-modal network comprises two separate hashing networks for MS and PAN ship images. Our hashing networks leverage a feature semi-interaction module to effectively capture correlations between different modalities, thereby acquiring robust representations. These representations are subsequently transformed into common hash codes using the semantic ranking objective function. These hash codes are used to construct hash tables for each modality. The objective function includes a shared semantic alignment loss and a margin-free ranking loss. The alignment loss employs a shared semantic layer to preserve label-level semantic similarity among cross-modal ship images, while the ranking loss incorporates hard examples to effectively learn complex similarity ranking relationships. Extensive experiments are conducted on cross-modal RS ship image datasets to evaluate the retrieval performance of FSISR. These results consistently demonstrate that FSISR outperforms state-of-the-art methods in cross-modal RS ship image retrieval. The proposed method

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studies cross-modal hashing methods on RS ship images from two satellite sources. EO data archives contain multiple types of satellite sources. In future work, we may extend the proposed FSISR method to multiple satellite sources.

REFERENCES

[1] Y. Li, J. Ma, and Y. Zhang, "Image retrieval from remote sensing big data: A survey," *Inf. Fusion*, vol. 67, pp. 94–115, Mar. 2021.

[2] W. Zhou, H. Guan, Z. Li, Z. Shao, and M. R. Delavar, "Remote sensing image retrieval in the past decade: Achievements, challenges, and future directions," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1447–1473, 2023.

[3] J. Kang et al., "DisOptNet: Distilling semantic knowledge from optical images for weather-independent building segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4706315.

[4] J. Kang, Z. Wang, R. Zhu, X. Sun, R. Fernandez-Beltran, and A. Plaza, "PiCoCo: Pixelwise contrast and consistency learning for semisupervised building footprint segmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 10548–10559, 2021.

[5] W. Xiong, Z. Xiong, Y. Cui, L. Huang, and R. Yang, "An interpretable fusion Siamese network for multi-modality remote sensing ship image retrieval," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 6, pp. 2696–2712, 2022.

[6] R. Hu, J. Yang, B. Zhu, and Z. Guo, "Research on ship image retrieval based on BoVW model under Hadoop platform," in *Proc. Int. Conf. Inf. Sci. Syst.*, Apr. 2018, pp. 156–160.

[7] W. Xiong, Y. Lv, X. Zhang, and Y. Cui, "Learning to translate for cross-source remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4860–4874, Jul. 2020.

[8] F. Xu, W. Yang, T. Jiang, S. Lin, H. Luo, and G.-S. Xia, "Mental retrieval of remote sensing images via adversarial sketch-image feature learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 7801–7814, Nov. 2020.

[9] Y. Fang, P. Li, J. Zhang, and P. Ren, "Cohesion intensive hash code book co-construction for efficiently localizing sketch depicted scenes," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5629016.

[10] G. Sumbul, M. Ravanbakhsh, and B. Demir, "Informative and representative triplet selection for multilabel remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5405811.

[11] U. Chaudhuri, B. Banerjee, A. Bhattacharya, and M. Datcu, "A zero-shot sketch-based intermodal object retrieval scheme for remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[12] N.-C. Ristea, A. Anghel, M. Datcu, and B. Chapron, "Guided unsupervised learning by subaperture decomposition for ocean SAR image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5207111.

[13] S. Wang, D. Hou, and H. Xing, "A self-supervised-driven open-set unsupervised domain adaptation method for optical remote sensing image scene classification and retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5605515.

[14] Y. Chen, S. Xiong, L. Mou, and X. X. Zhu, "Deep quadruple-based hashing for remote sensing image-sound retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.

[15] Y. Chen, X. Lu, and S. Wang, "Deep cross-modal image-voice retrieval in remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7049–7061, Oct. 2020.

[16] H. Ning, B. Zhao, and Y. Yuan, "Semantics-consistent representation learning for remote sensing image–voice retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4700614.

[17] J. Huang, Y. Chen, S. Xiong, and X. Lu, "Self-supervision interactive alignment for remote sensing image–audio retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4701114.

[18] Y. Chen, J. Huang, S. Xiong, and X. Lu, "Fine aligned discriminative hashing for remote sensing image-audio retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4702012.

[19] Z. Yuan et al., "Exploring a fine-grained multiscale method for cross-modal remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4404119.

[20] Z. Yuan et al., "Remote sensing cross-modal text-image retrieval based on global and local information," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5620616.

[21] Q. Cheng, Y. Zhou, P. Fu, Y. Xu, and L. Zhang, "A deep semantic alignment network for the cross-modal image-text retrieval in remote sensing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4284–4297, 2021.

[22] H. Yu et al., "Text-image matching for cross-modal remote sensing image retrieval via graph neural network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 812–824, 2023.

[23] X. Tang, Y. Wang, J. Ma, X. Zhang, F. Liu, and L. Jiao, "Interacting-enhancing feature transformer for cross-modal remote sensing image and text retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5611715.

[24] Y. Li, Y. Zhang, X. Huang, and J. Ma, "Learning source-invariant deep hashing convolutional neural networks for cross-source remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 11, pp. 6521–6536, Nov. 2018.

[25] W. Xiong, Z. Xiong, Y. Zhang, Y. Cui, and X. Gu, "A deep cross-modality hashing network for SAR and optical remote sensing images retrieval," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 5284–5296, 2020.

[26] Y. Li, Y. Zhang, X. Huang, H. Zhu, and J. Ma, "Large-scale remote sensing image retrieval by deep hashing neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 950–965, Feb. 2018.

[27] P. Li et al., "Hashing nets for hashing: A quantized deep learning to hash framework for remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7331–7345, Oct. 2020.

[28] W. Song, Z. Gao, R. Dian, P. Ghamisi, Y. Zhang, and J. A. Benediktsson, "Asymmetric hash code learning for remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5617514.

[29] L. Han et al., "Central cohesion gradual hashing for remote sensing image retrieval," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[30] M. Huang, L. Dong, W. Dong, and G. Shi, "Supervised contrastive learning based on fusion of global and local features for remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5208513.

[31] B. Demir and L. Bruzzone, "Hashing-based scalable remote sensing image search and retrieval in large archives," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 2, pp. 892–904, Feb. 2016.

[32] C. Liu, J. Ma, X. Tang, F. Liu, X. Zhang, and L. Jiao, "Deep hash learning for remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3420–3443, Apr. 2021.

[33] J. Kang, R. Fernandez-Beltran, Z. Ye, X. Tong, and A. Plaza, "Deep hashing based on class-discriminated neighborhood embedding," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 5998–6007, 2020.

[34] L. Han, P. Li, X. Bai, C. Grecos, X. Zhang, and P. Ren, "Cohesion intensive deep hashing for remote sensing image retrieval," *Remote Sens.*, vol. 12, no. 1, p. 101, Dec. 2019.

[35] R. Fernandez-Beltran, B. Demir, F. Pla, and A. Plaza, "Unsupervised remote sensing image retrieval using probabilistic latent semantic hashing," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 2, pp. 256–260, Feb. 2021.

[36] W. Song, S. Li, and J. A. Benediktsson, "Deep hashing learning for visual and semantic retrieval of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9661–9672, Nov. 2021.

[37] P. Liu, Z. Liu, X. Shan, and Q. Zhou, "Deep hash remote-sensing image retrieval assisted by semantic cues," *Remote Sens.*, vol. 14, no. 24, p. 6358, Dec. 2022.

[38] L. Han, P. Li, A. Plaza, and P. Ren, "Hashing for localization (HfL): A baseline for fast localizing objects in a large-scale scene," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5609916.

[39] Y. Chen, J. Huang, L. Mou, P. Jin, S. Xiong, and X. X. Zhu, "Deep saliency smoothing hashing for drone image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4700913.

[40] J. Han, P. Li, Y. Tao, and P. Ren, "Encrypting hashing against localization," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5607414.

[41] X. Tang et al., "Meta-hashing for remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5615419.

[42] D. Hou, S. Wang, X. Tian, and H. Xing, "PCLUDA: A pseudo-label consistency learning-based unsupervised domain adaptation method for cross-domain optical remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5600314.

[43] Y. Sun et al., "Unsupervised deep hashing through learning soft pseudo label for remote sensing image retrieval," *Knowl.-Based Syst.*, vol. 239, Mar. 2022, Art. no. 107807.

[44] T. Zhang et al., "FFN: Fountain fusion net for arbitrary-oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5609913.

[45] L. Gao, D. Wang, L. Zhuang, X. Sun, M. Huang, and A. Plaza, "BS3LNet: A new blind-spot self-supervised learning network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5504218.

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[46] L. Zhuang, M. K. Ng, L. Gao, J. Michalski, and Z. Wang, "Eigen-image2Eigenimage (E2E): A self-supervised deep learning network for hyperspectral image denoising," *IEEE Trans. Neural Netw. Learn. Syst.*, pp. 1–15, Jul. 2023.
[47] E. Hussain and J. Shan, "Mapping major floods with optical and SAR satellite images," in *Proc. IEEE Geosci. Remote Sens. Symp.*, Jul. 2010, pp. 25–30.
[48] J. Kang, T. Ji, Z. Zhang, and R. Fernandez-Beltran, "SAR time series despeckling via nonlocal matrix decomposition in logarithm domain," *Signal Process.*, vol. 209, Aug. 2023, Art. no. 109040.
[49] J. Song, Y. Yang, Y. Yang, Z. Huang, and H. T. Shen, "Inter-media hashing for large-scale retrieval from heterogeneous data sources," in *Proc. ACM SIGMOD Int. Conf. Manage. Data*, Jun. 2013, pp. 785–796.
[50] J. Zhou, G. Ding, and Y. Guo, "Latent semantic sparse hashing for cross-modal similarity search," in *Proc. Int. Conf. Res. Develop. Inf. Retr.*, 2014, pp. 415–424.
[51] G. Ding, Y. Guo, and J. Zhou, "Collective matrix factorization hashing for multimodal data," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 2083–2090.
[52] D. Zhang and W.-J. Li, "Large-scale supervised multimodal hashing with semantic correlation maximization," in *Proc. AAAI Conf. Artif. Intell.*, vol. 28, 2014, pp. 2177–2183.
[53] Q.-Y. Jiang and W.-J. Li, "Deep cross-modal hashing," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 3232–3240.
[54] Y. Cao, B. Liu, M. Long, and J. Wang, "Cross-modal Hamming hashing," in *Proc. Eur. Conf. Comput. Vis.*, in Lecture Notes in Computer Science, vol. 11205, 2018, pp. 207–223.
[55] L. Jin, Z. Li, and J. Tang, "Deep semantic multimodal hashing network for scalable image-text and video-text retrievals," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 4, pp. 1838–1851, Apr. 2023.
[56] Y. Fang, P. Li, J. Zhang, and P. Ren, "Cohesion intensive hash code book coconstruction for efficiently localizing sketch depicted scenes," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5629016.
[57] Y. Sun et al., "Multisensor fusion and explicit semantic preserving-based deep hashing for cross-modal remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5219614.
[58] U. Chaudhuri, R. Bose, B. Banerjee, A. Bhattacharya, and M. Datcu, "Zero-shot cross-modal retrieval for remote sensing images with minimal supervision," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4708615.
[59] H. Zhu, M. Long, J. Wang, and Y. Cao, "Deep hashing network for efficient similarity retrieval," in *Proc. AAAI Conf. Artif. Intell.*, 2016, pp. 2415–2421.
[60] M. M. Zhang, J. Choi, K. Daniilidis, M. T. Wolf, and C. Kanan, "VAIS: A dataset for recognizing maritime imagery in the visible and infrared spectrums," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2015, pp. 10–16.
[61] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 8026–8037.
[62] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.



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